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9.Network Layer: IP addressing

Computer Networks



CS4054 - 2026II
PROF.: CWILLIAMS@UTEC.EDU.PE, JGONZALEZ@UTEC.EDU.PE
SRC: HARRIS & HARRIS - DIGITAL DESIGN AND COMPUTER ARCHITECTURE RISC-V

Executive Summary

- **Motivation: Routers are fundamental components of network architectures, enabling communication across interconnected networks.**
- **Problem: How do routers provide packet forwarding and IP addressing in large-scale networks?**
- **Overview:**

- **Internet Protocol (IP) addressing fundamentals.**
- **Router operation and packet forwarding.**
- **Subnetworks and hierarchical addressing.**
- **Dynamic IP address assignment using DHCP.**

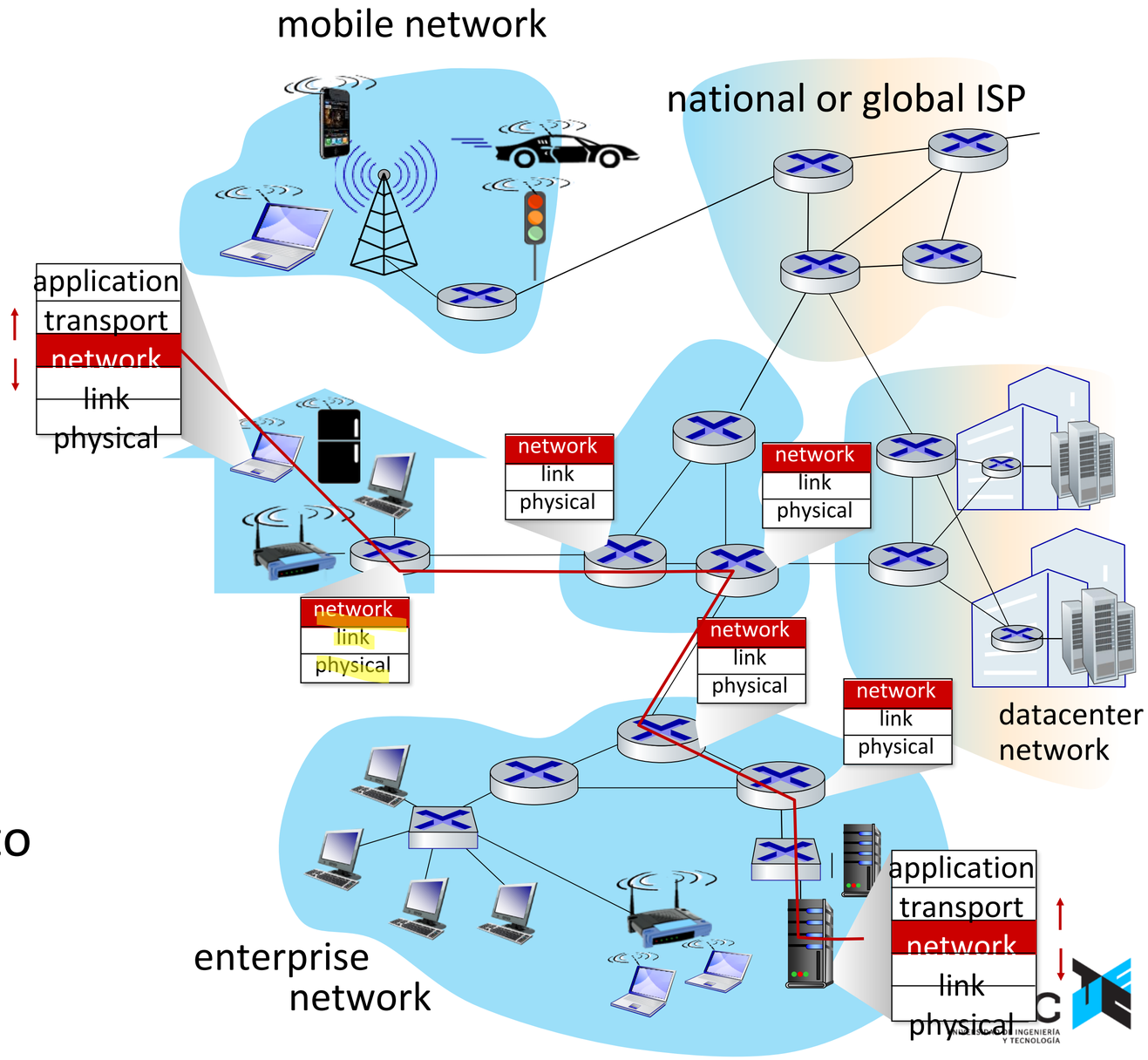
- **Conclusion: We introduced the Internet Protocol and studied how routers, hierarchical addressing, subnetting, and DHCP enable scalable network communication.**

Outline

- Introduction
- Addressing IPv4
- DHCP and Hierarchical Addressing
- Conclusion

Network-layer services and protocols

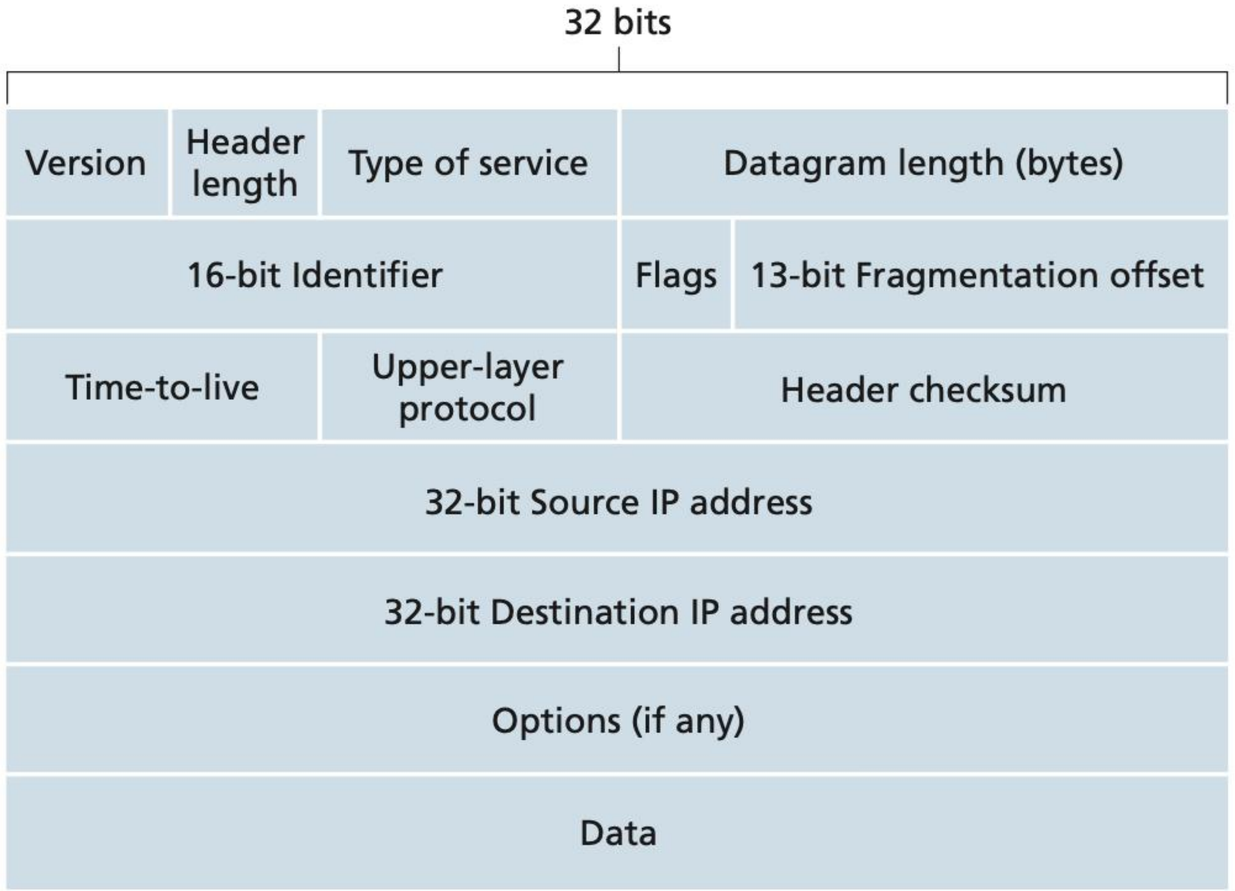
- transport segment from sending to receiving host
  - sender: encapsulates segments into datagrams, passes to link layer
  - receiver: delivers segments to transport layer protocol
- network layer protocols in *every Internet device*: hosts, routers
- routers:
  - examines header fields in all IP datagrams passing through it
  - moves datagrams from input ports to output ports to transfer datagrams along end-end path



IP Datagram

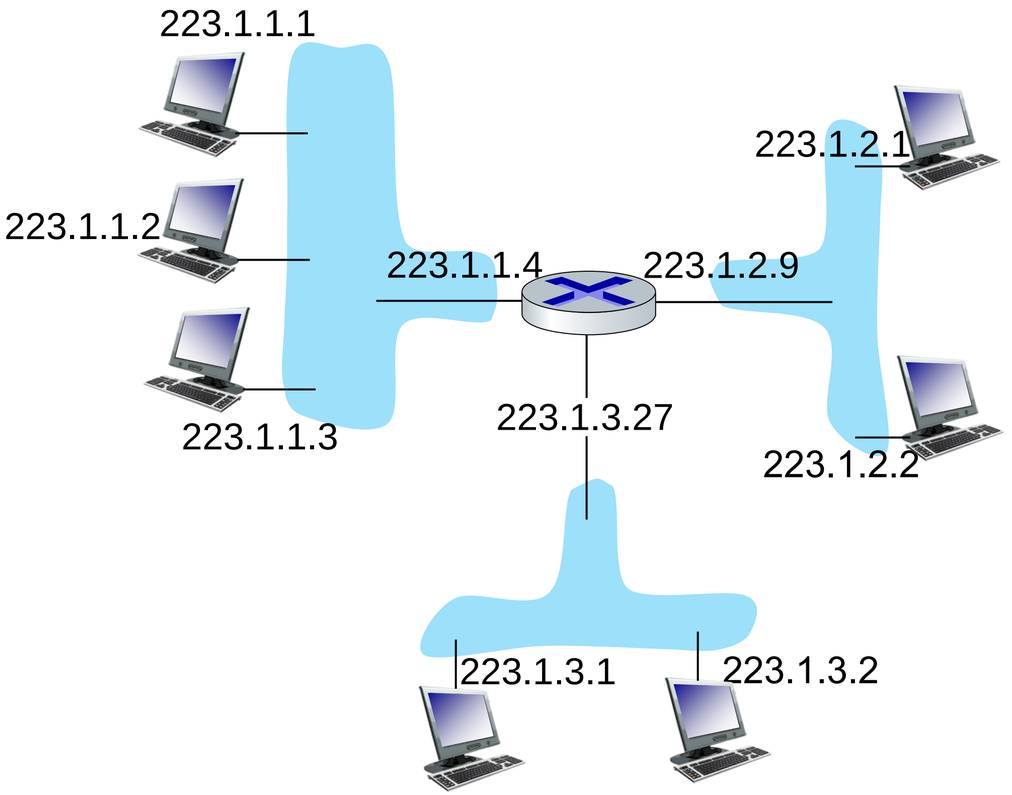
IP Overhead

- 20 bytes from TCP
- 20 bytes from IP
- = 40 bytes + overhead from application layer



IP addressing: Introduction

- IP address: 32-bit identifier associated with each host or router *interface*
- interface: connection between host/router and physical link
  - router's typically have multiple interfaces
  - host typically has one or two interfaces (e.g., wired Ethernet, wireless 802.11)



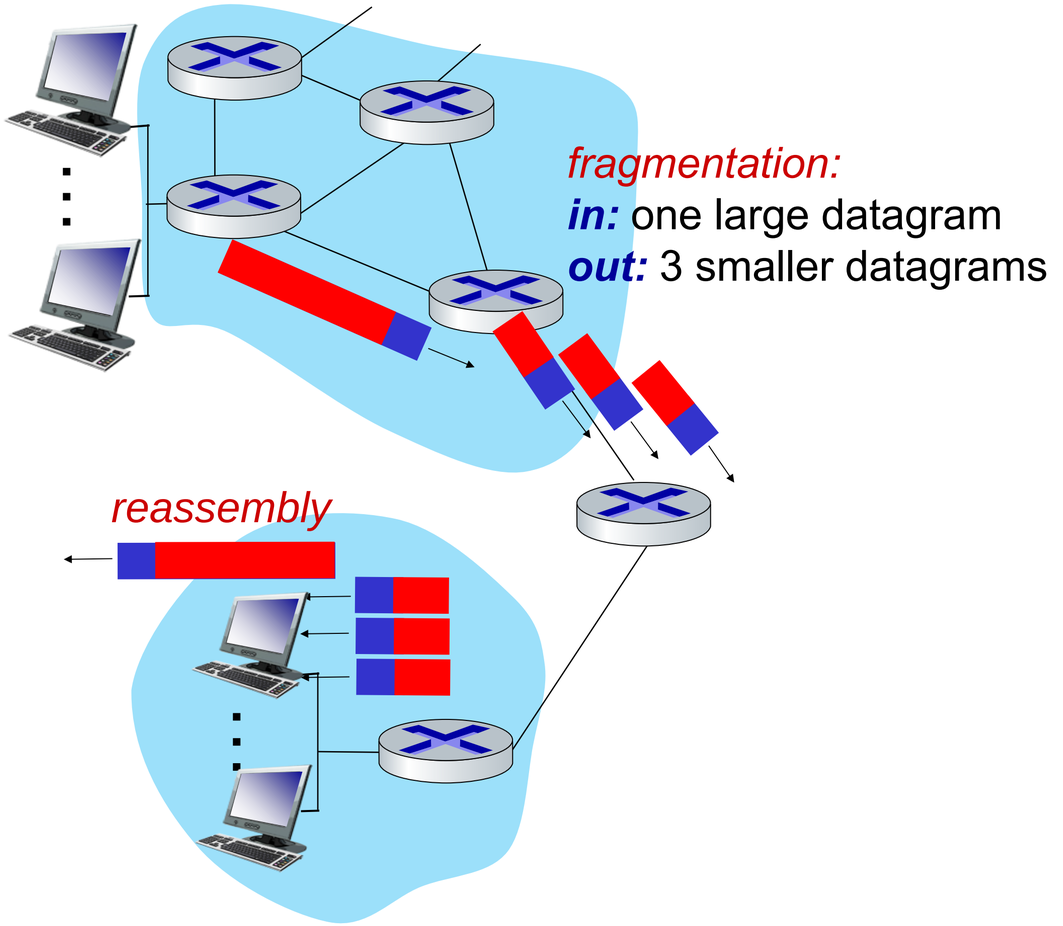
dotted-decimal IP address notation:

223.1.1.1 = 11011111 00000001 00000001 00000001

223 1 1 1

IP fragmentation/reassembly

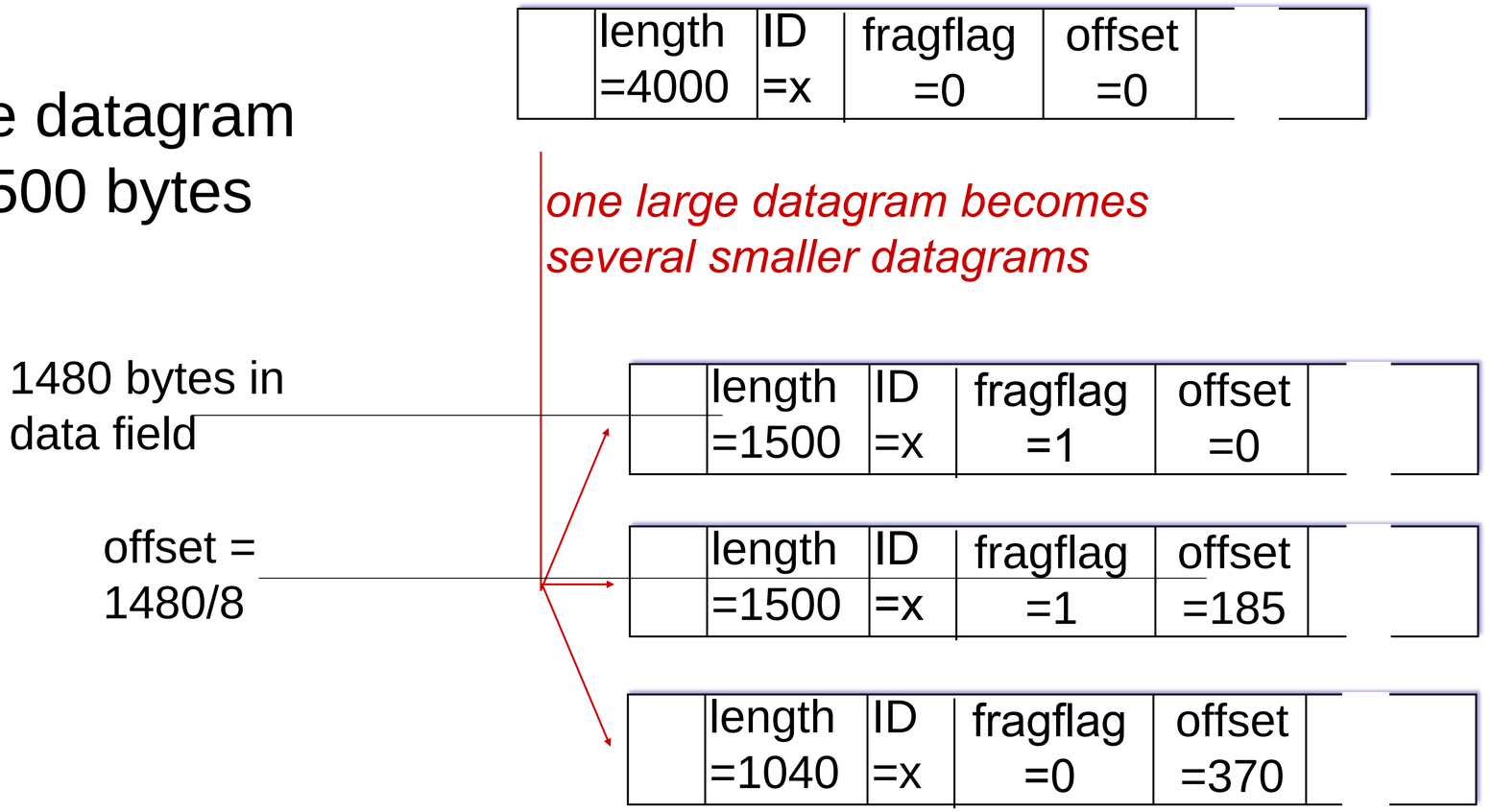
- network links have MTU (max. transfer size) - largest possible link-level frame
  - different link types, different MTUs
- large IP datagram divided (“fragmented”) within net
  - one datagram becomes several datagrams
  - “reassembled” only at *destination*
  - IP header bits used to identify, order related fragments



IP fragmentation/reassembly

example:

- 4000 byte datagram
- MTU = 1500 bytes



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Two key network-layer functions

network-layer functions:

- *forwarding:* move packets from a router's input link to appropriate router output link
  - *routing:* determine route taken by packets from source to destination
    - *routing algorithms*

analogy: taking a trip

- *forwarding:* process of getting through single interchange
  - *routing:* process of planning trip from source to destination

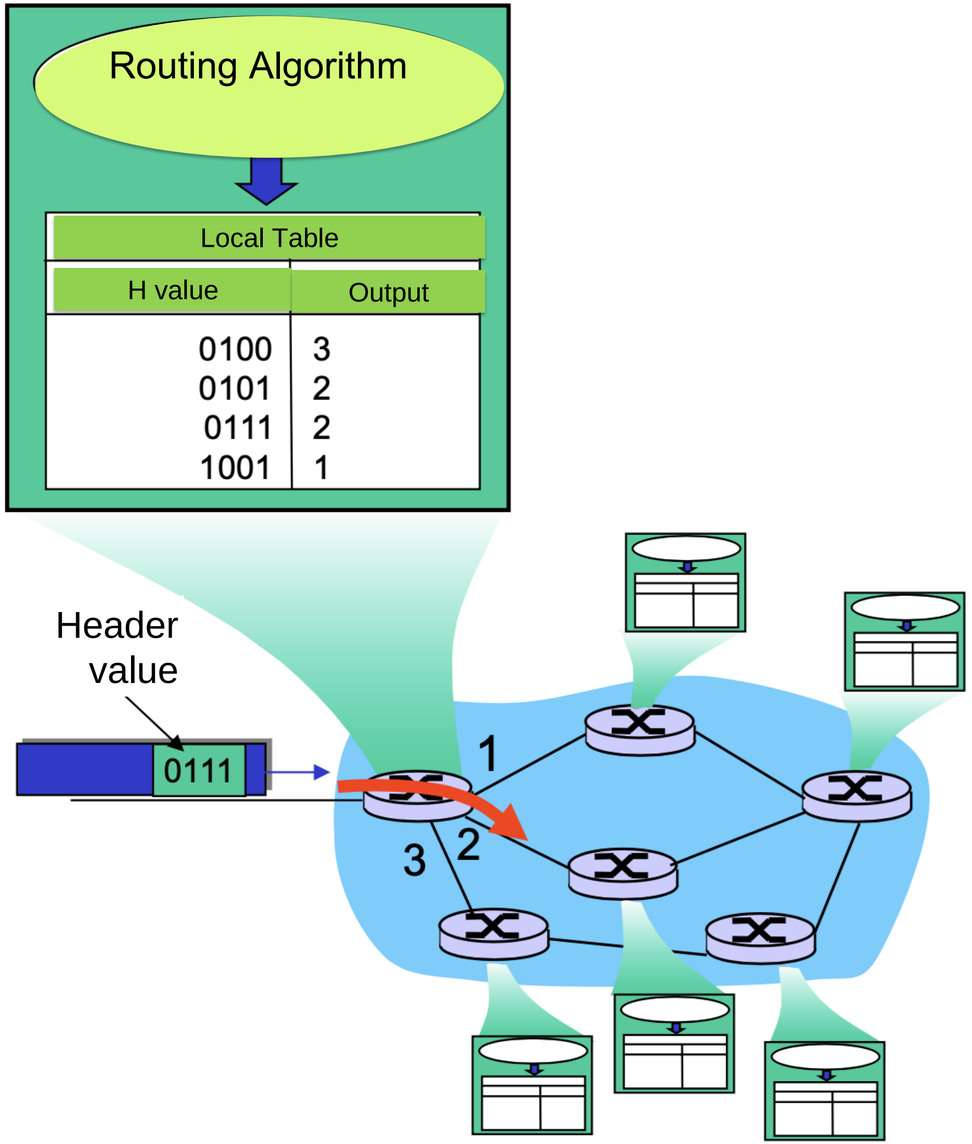


forwarding



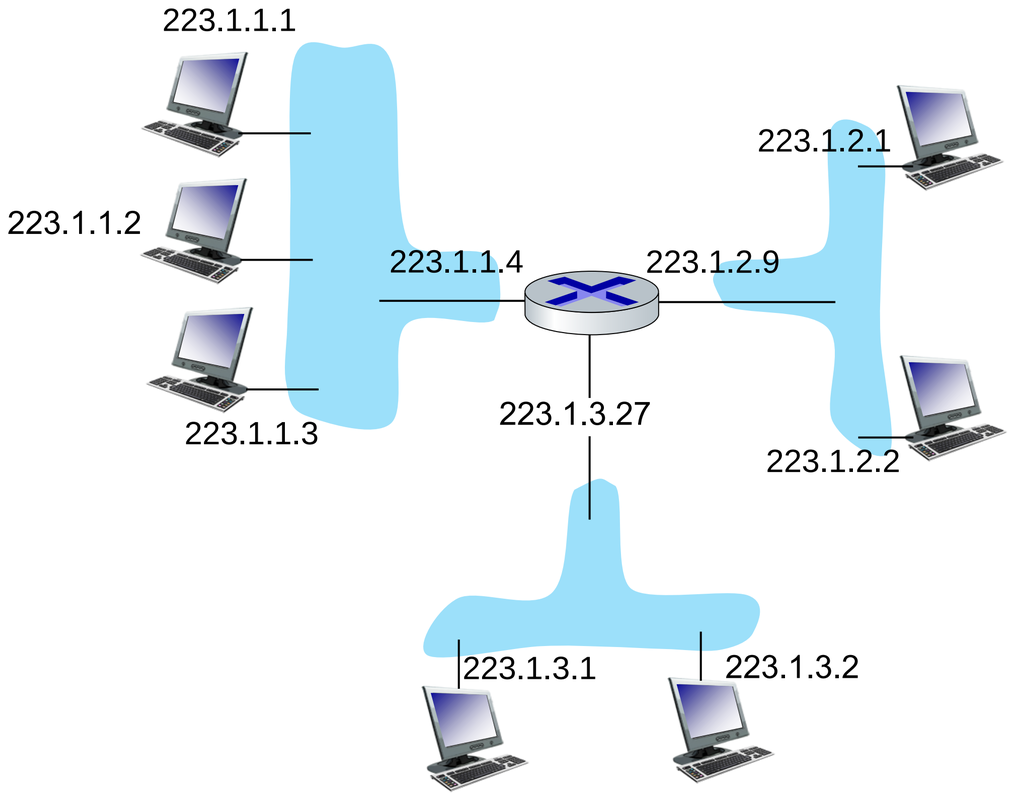
routing

Two key network-layer functions



IP addressing: introduction

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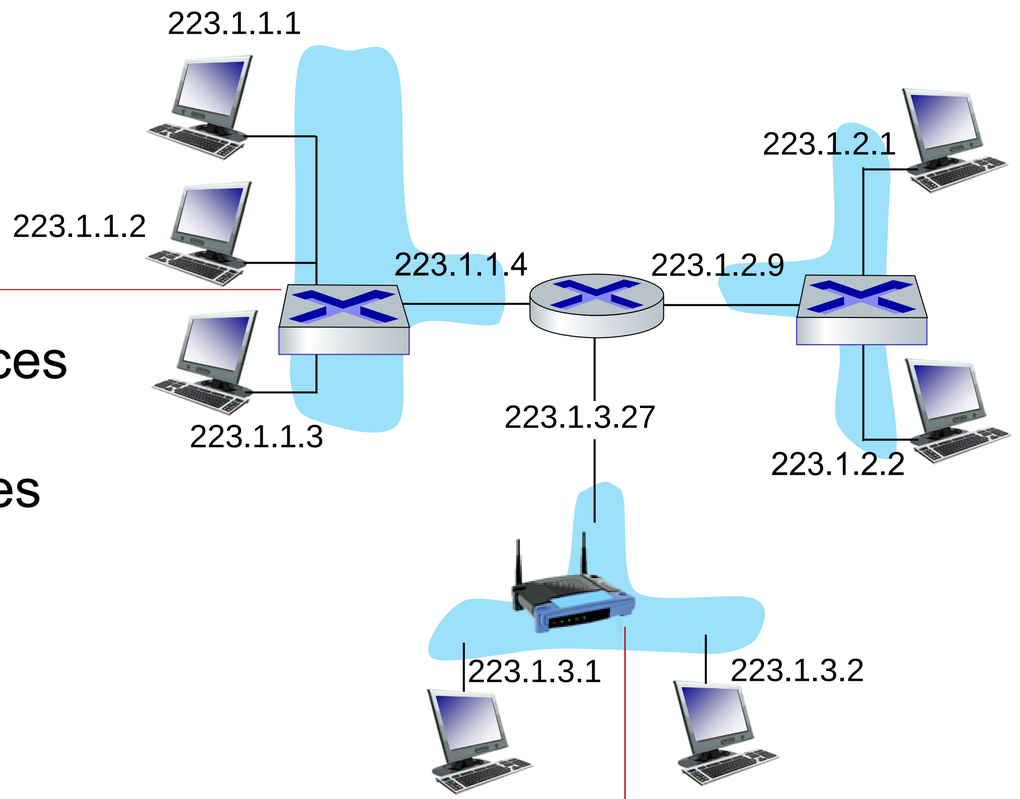
223 1 1 1

IP addressing: introduction

Q: how are interfaces actually connected?

A: we'll learn about that in Kurose book (chapters 6, 7)

For now: don't need to worry about how one interface is connected to another (with no intervening router)



A: wired Ethernet interfaces connected by Ethernet switches

A: wireless WiFi interfaces connected by WiFi base station

IP addressing: CIDR

CIDR: Classless InterDomain Routing (pronounced “cider”)

- subnet portion of address of arbitrary length
- address format: a.b.c.d/x, where x is # bits in subnet portion of address



200.23.16.0/23

Special Addresses

- Loop Back
  127.0.0.1(localhost)
  Local communication interface
- Global broadcast
  255.255.255.255(all 1s)
- Sub-network broadcast
  Last sub-network address
  Mask (OR lógico) IP
- Sub-network address
  First address in the

Mask (AND lógico) IP

IP routing: Netstat command

- IP routing table:
- Try options -nr

```
IPv4 Route Table
===========================================================================
Active Routes:
Network Destination        Netmask          Gateway       Interface  Metric
          0.0.0.0          0.0.0.0    192.168.100.1    192.168.100.8     30
        127.0.0.0        255.0.0.0         On-link         127.0.0.1    331
        127.0.0.1  255.255.255.255         On-link         127.0.0.1    331
  127.255.255.255  255.255.255.255         On-link         127.0.0.1    331
    192.168.100.0    255.255.255.0         On-link     192.168.100.8    286
    192.168.100.8  255.255.255.255         On-link     192.168.100.8    286
  192.168.100.255  255.255.255.255         On-link     192.168.100.8    286
        224.0.0.0        240.0.0.0         On-link         127.0.0.1    331
        224.0.0.0        240.0.0.0         On-link     192.168.100.8    286
  255.255.255.255  255.255.255.255         On-link         127.0.0.1    331
  255.255.255.255  255.255.255.255         On-link     192.168.100.8    286
===========================================================================
```

IP routing: Ifconfig command

- Ifconfig and network adapters

```
Windows IP Configuration


Ethernet adapter Ethernet 3:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter WiFi:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::7b94:1f7a:e5d5:314c%15
   IPv4 Address. . . . . . . . . . . : 192.168.100.8
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 192.168.100.1

Ethernet adapter Bluetooth Network Connection:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Ethernet adapter Ethernet 2:

   Media State . . . . . . . . . . . : Media disconnected
```

IP routing: traceroute

- Traceroute command shows network layer routing

```
C:\Users\carlo>tracert google.com

Tracing route to google.com [142.251.132.174]
over a maximum of 30 hops:

  1     2 ms     4 ms     3 ms  192.168.100.1
  2    18 ms     8 ms     5 ms  10.190.0.1
  3     5 ms     4 ms     5 ms  10.10.7.21
  4    10 ms     3 ms     6 ms  10.10.7.22
  5     5 ms     4 ms     6 ms  38.253.184.11
  6     5 ms     5 ms     5 ms  181.177.224.17
  7    10 ms     4 ms     4 ms  213.140.50.252
  8    12 ms     7 ms     5 ms  84.16.12.79
  9     *       40 ms     *     94.142.99.220
 10    40 ms    40 ms    40 ms  176.52.252.37
 11    43 ms    42 ms    42 ms  64.233.174.147
 12    42 ms    42 ms    48 ms  192.178.104.56
 13    44 ms    44 ms    43 ms  142.250.215.221
 14   103 ms   102 ms   102 ms  216.239.63.0
 15   159 ms   165 ms   159 ms  142.251.248.26
 16   172 ms   172 ms   172 ms  142.250.225.53
 17   170 ms   171 ms   171 ms  142.251.232.243
 18   169 ms   169 ms   169 ms  142.250.210.135
 19   169 ms   168 ms   168 ms  ncboga-ak-in-f14.1e100.net [142.251.132.174]

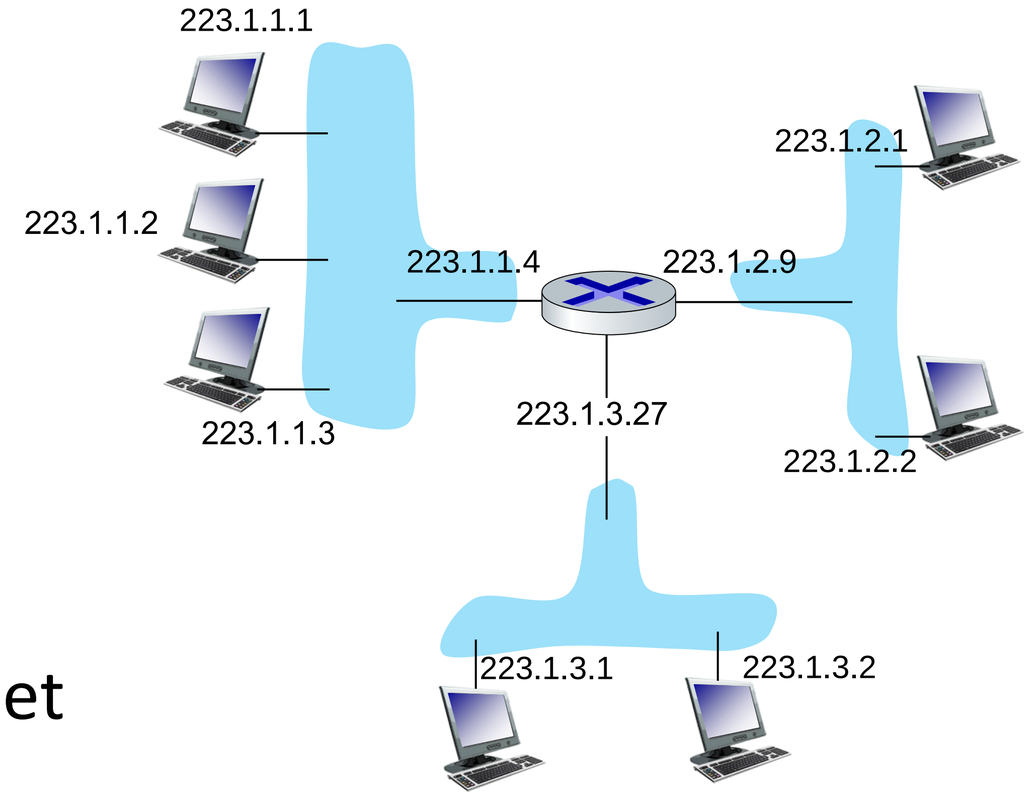
Trace complete.
```

MAC and ARP

- 32 bit – IP address:
  - Network layer addresses
  - Used to route the datagram to its destination (sub-network)
- MAC Address (or LAN or physical layer or Ethernet - eth) :
  - function: “take frame” from one interface to another physically connected interface (at same network)
  - 48 bit MAC Adresses (most of LANs) written in ROM of NIC, can be also configurable in software

Subnets

- *What's a subnet ?*
  - device interfaces that can physically reach each other without passing through an intervening router
- IP addresses have structure:
  - subnet part: devices in same subnet have common high order bits
  - host part: remaining low order bits

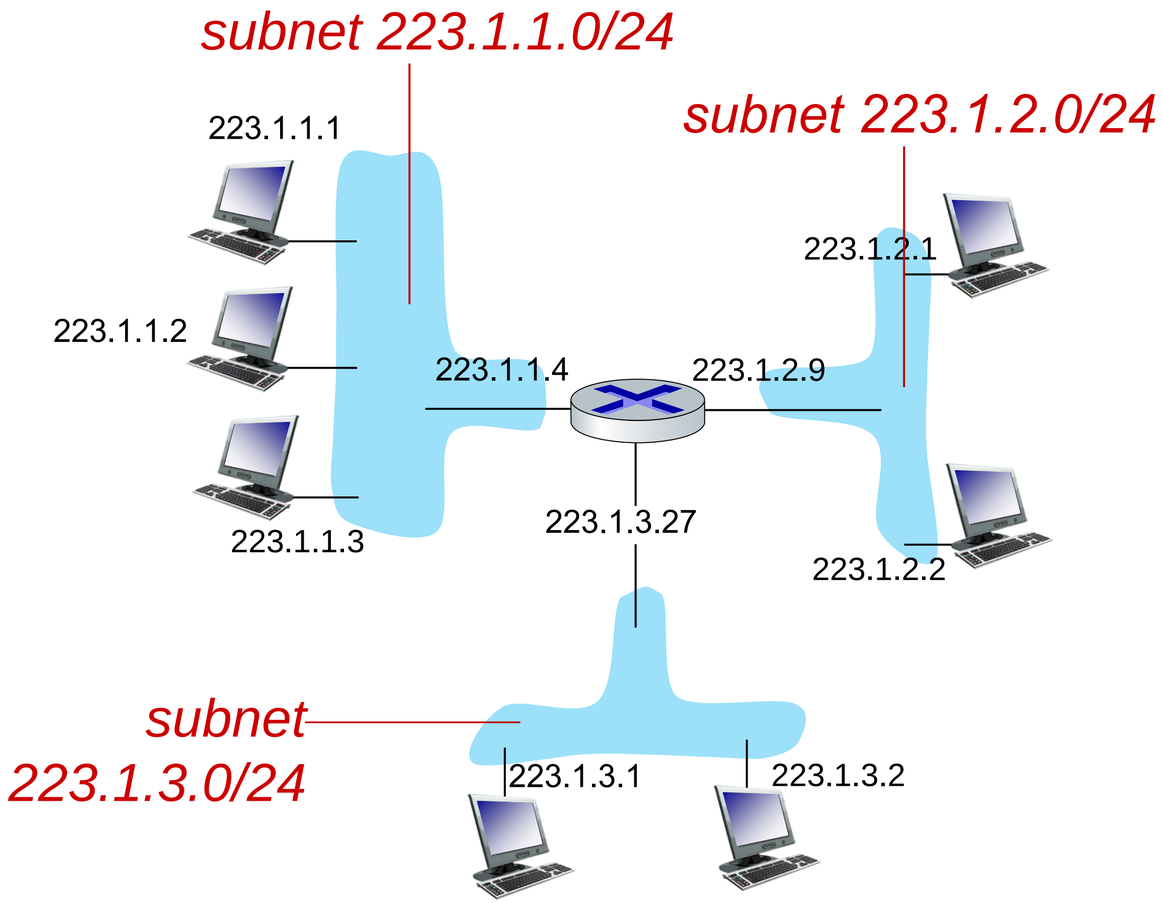


network consisting of 3 subnets

Subnets

Recipe for defining subnets:

- detach each interface from its host or router, creating "islands" of isolated networks
- each isolated network is called a *subnet*

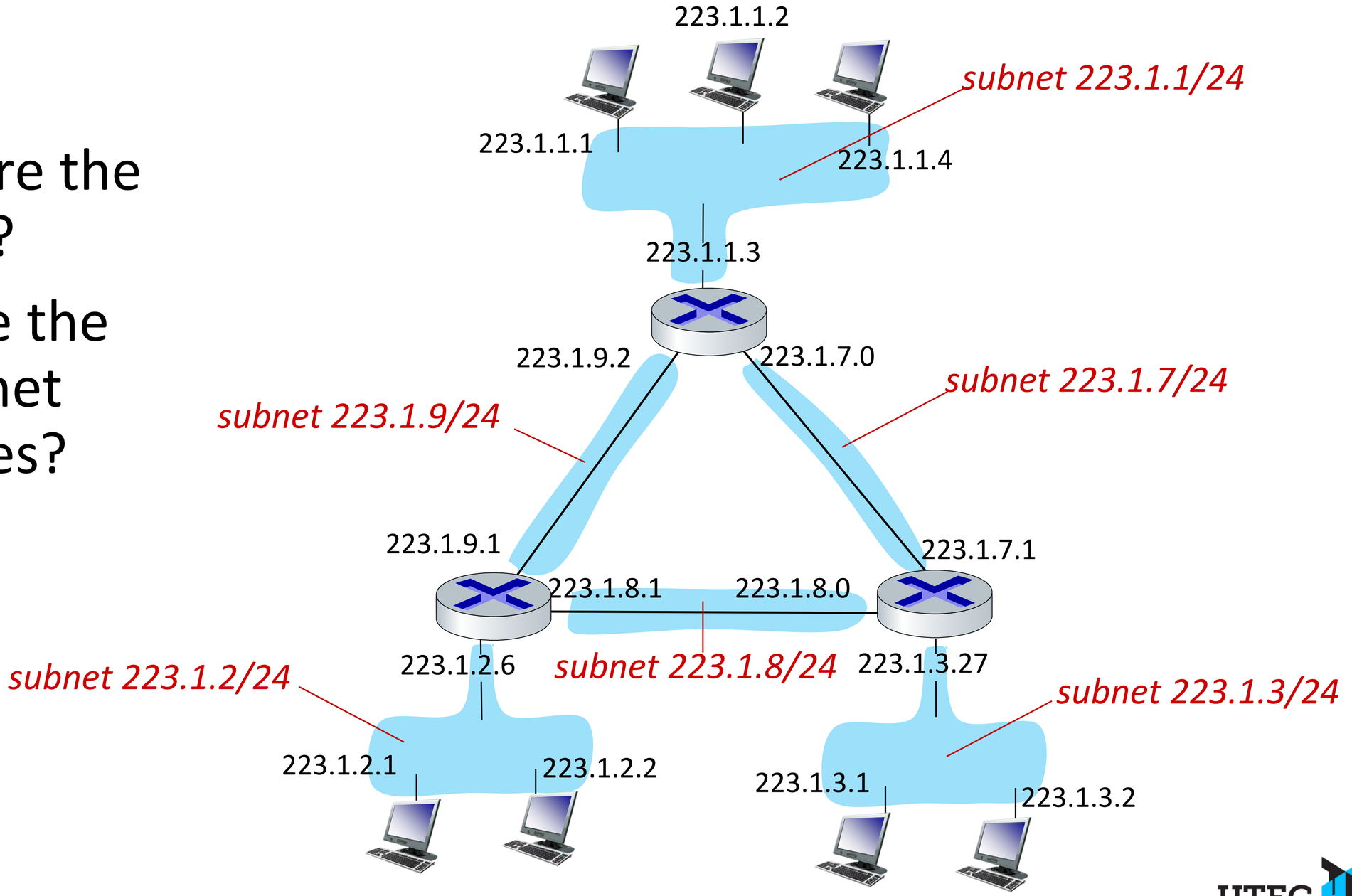


subnet mask: /24

(high-order 24 bits: subnet part of IP address)

Subnets

- where are the subnets?
- what are the /24 subnet addresses?



Outline

Introduction

Addressing IPv4

DHCP and Hierarchical Addressing

Conclusion

IP addresses: how to get one?

That's actually two questions:

1. Q: How does a *host* get IP address within its network (host part of address)?
2. Q: How does a *network* get IP address for itself (network part of address)

How does *host* get IP address?

- hard-coded by sysadmin in config file (e.g., /etc/rc.config in UNIX)
- DHCP: Dynamic Host Configuration Protocol: dynamically get address from as server
  - "plug-and-play"

DHCP: Dynamic Host Configuration Protocol

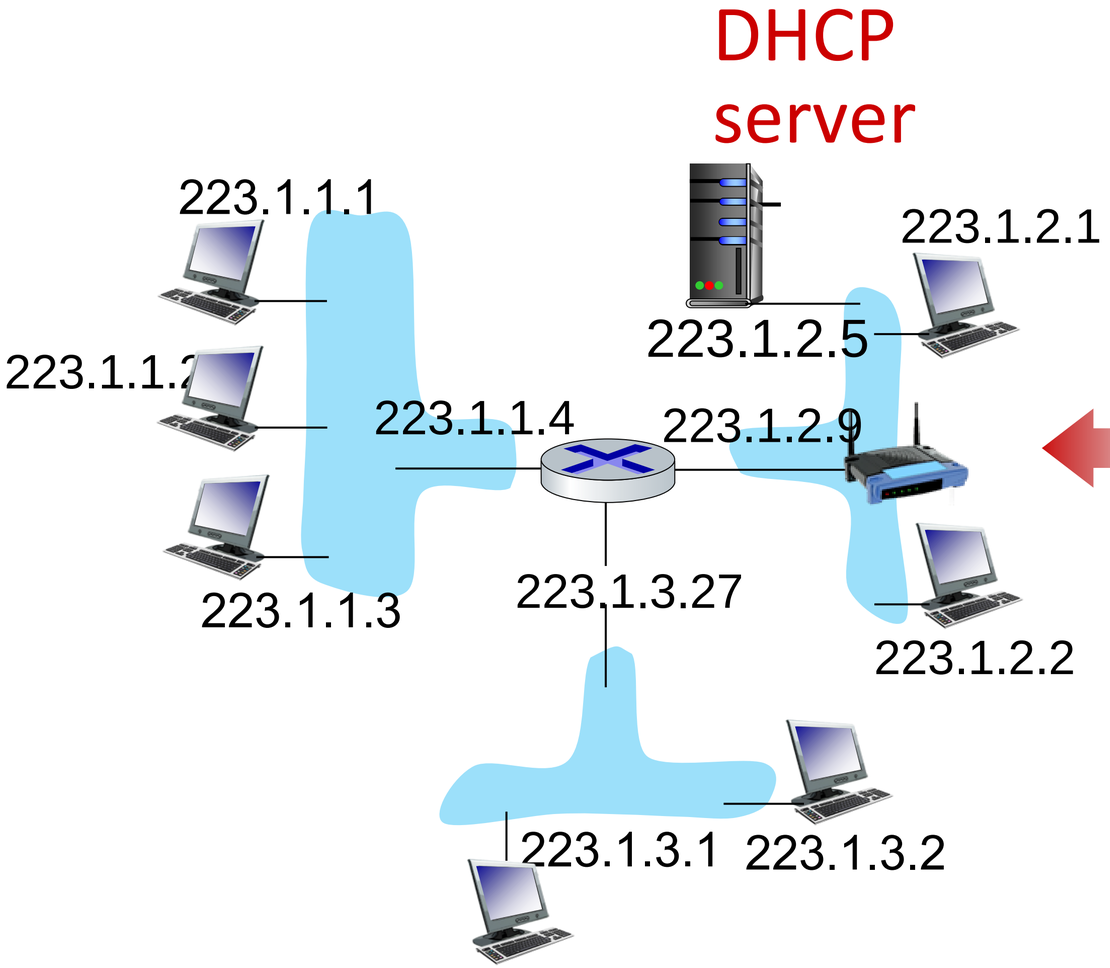
goal: host *dynamically* obtains IP address from network server when it "joins" network

- can renew its lease on address in use
- allows reuse of addresses (only hold address while connected/on)
- support for mobile users who join/leave network

DHCP overview:

- host broadcasts DHCP discover msg [optional]
- DHCP server responds with DHCP offer msg [optional]
- host requests IP address: DHCP request msg
- DHCP server sends address: DHCP ack msg

DHCP client-server scenario

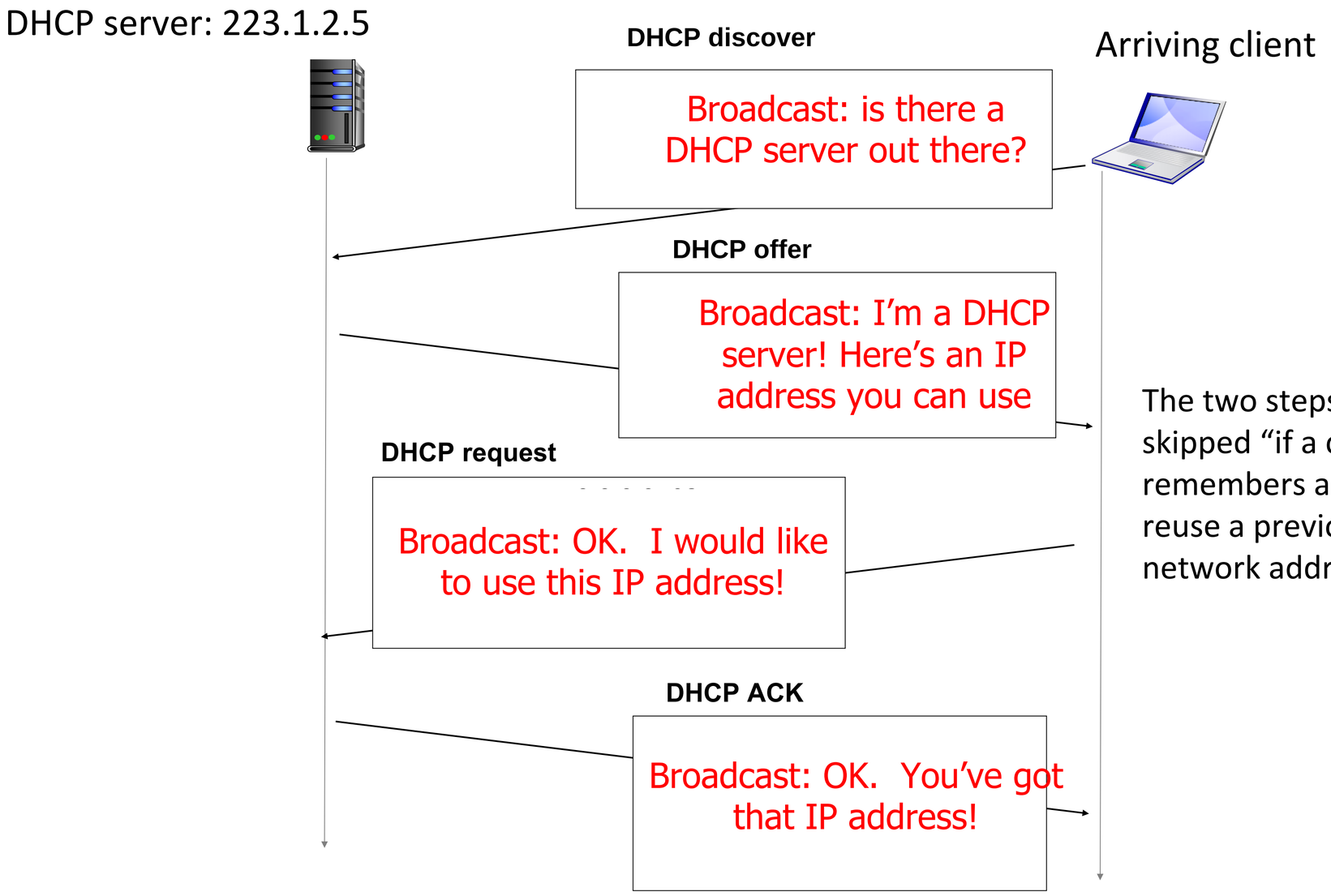


Typically, DHCP server will be co-located in router, serving all subnets to which router is attached



arriving DHCP client needs address in this network

DHCP client-server scenario



The two steps above can be skipped "if a client remembers and wishes to reuse a previously allocated network address" [RFC 2131]

DHCP: more than IP addresses

DHCP can return more than just allocated IP address on subnet:

- address of first-hop router for client
- name and IP address of DNS sever
- network mask (indicating network versus host portion of address)

IP addresses

Q: how does *network* get subnet part of IP address?

A: gets allocated portion of its provider ISP's address space

ISP's block	11001000 00010111 00010000	00000000	200.23.16.0/20

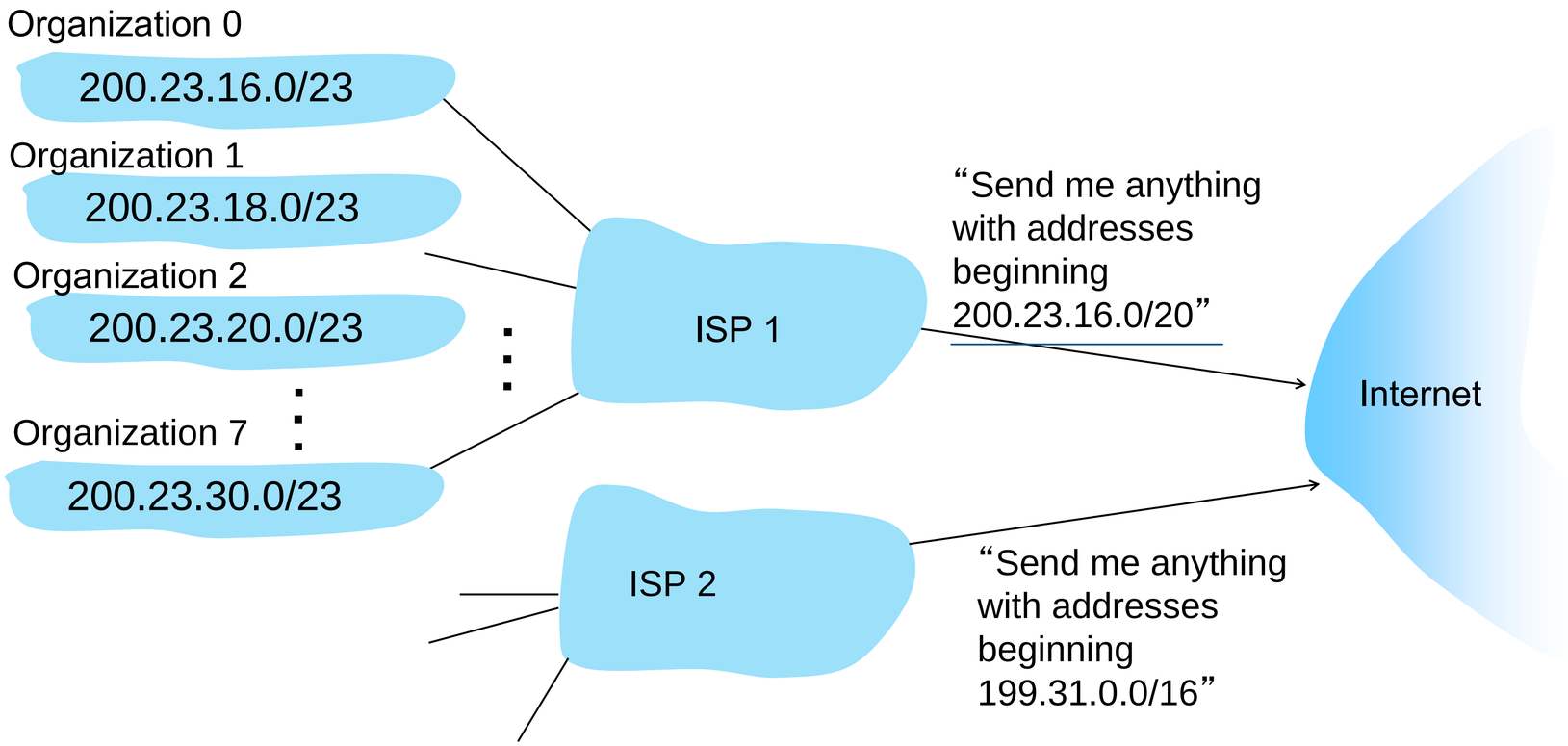
ISP can then allocate out its address space in 8 blocks:

Organization 0	11001000 00010111 00010000	00000000	200.23.16.0/23
Organization 1	11001000 00010111 00010010	00000000	200.23.18.0/23
Organization 2	11001000 00010111 00010100	00000000	200.23.20.0/23
...	.....	....	....
Organization 7	11001000 00010111 00011110	00000000	200.23.30.0/23

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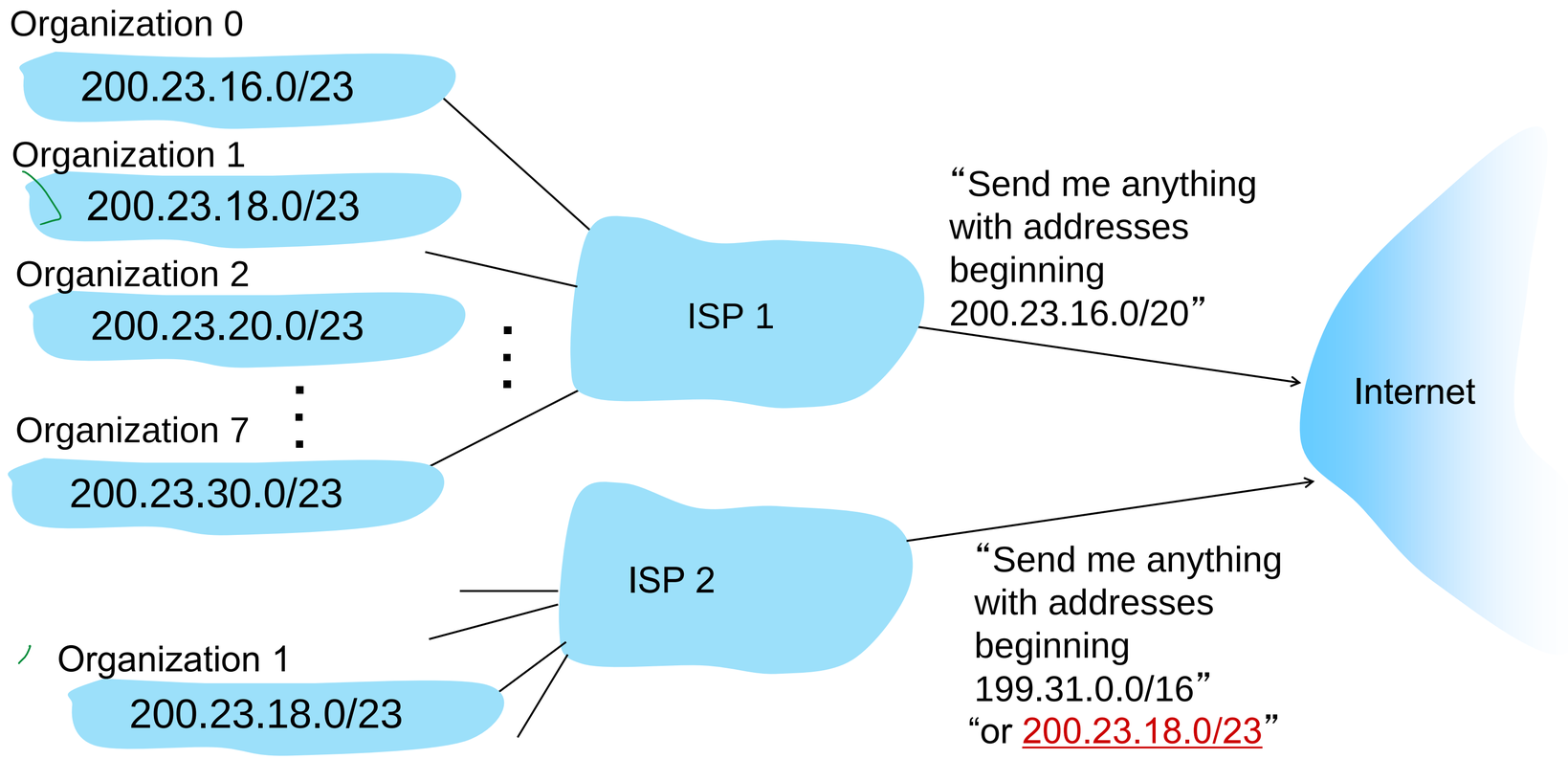
Hierarchical addressing: route aggregation

hierarchical addressing allows efficient advertisement of routing information:



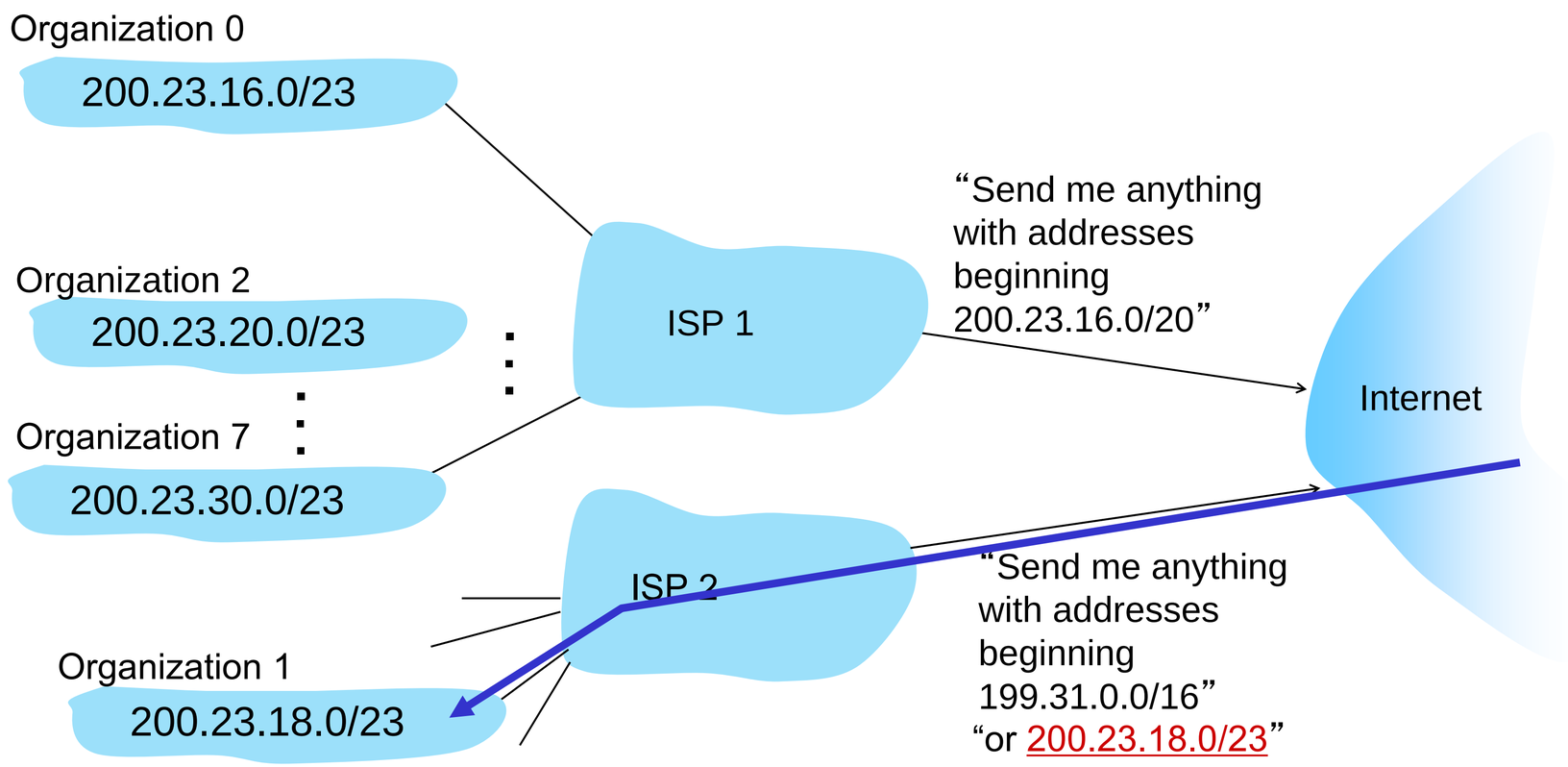
Hierarchical addressing: more specific routes

- Organization 1 moves from Fly-By-Night-ISP to ISPs-R-Us
- ISPs-R-Us now advertises a more specific route to Organization 1



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IP addressing: last words ...

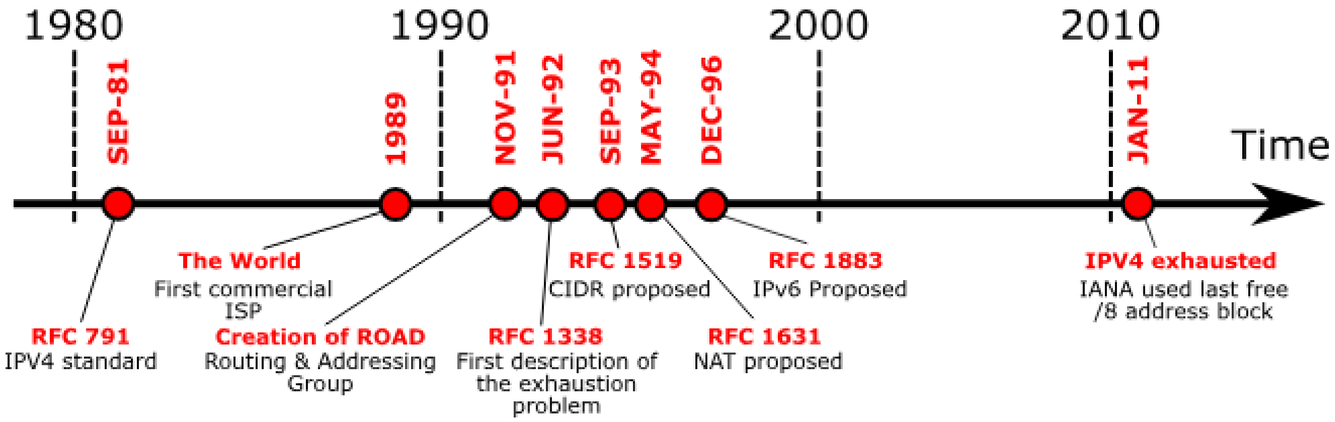
Q: how does an ISP get block of addresses?

A: ICANN: Internet Corporation for Assigned Names and Numbers http://www.icann.org/

- allocates IP addresses, through 5 regional registries (RRs) (who may then allocate to local registries)
- manages DNS root zone, including delegation of individual TLD (.com, .edu , …) management



IP addressing: last words ...





Vint Cerf

Q: are there enough 32-bit IP addresses?

- ICANN allocated last chunk of IPv4 addresses to RRs in 2011
- NAT (next) helps IPv4 address space exhaustion
- IPv6 has 128-bit address space

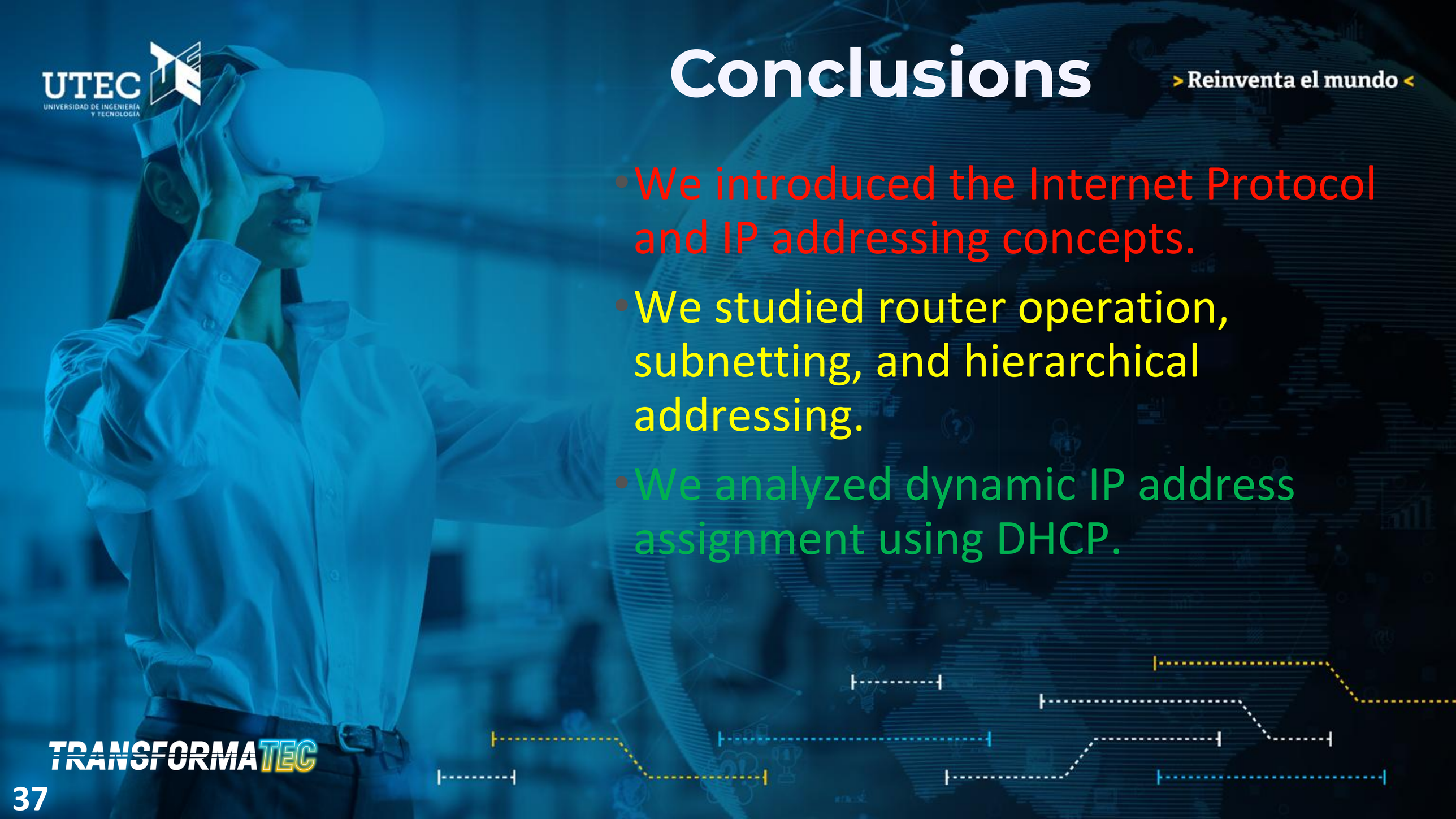
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Conclusions

> Reinventa el mundo <

- We introduced the Internet Protocol and IP addressing concepts.
- We studied router operation, subnetting, and hierarchical addressing.
- We analyzed dynamic IP address assignment using DHCP.

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